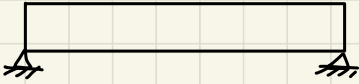


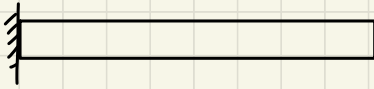
ESE 15 VIBRATION OF BEAMS

free undamped motion for different boundaries condition
(modal approach for multi DOF introducing external forces and springs)



Simply supported

analytical solution possible

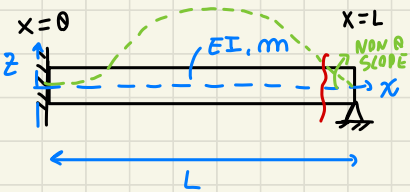


Cantilever beam

slender beam bending

↓ CANTILIVER BEAM ES 1

• With a pin on the right (additional constraints)



We expect ω_k, ϕ_k affected by new constraint

⇓ STIFFER

$\omega_k \approx \sqrt{\text{stiff}/\text{mass}}$ we expect $\omega_k \uparrow$

Using MATLAB script to solve the problem during PROBLEM 2 of EXAM

BOUNDRIES Condition

$x=0$: CLAMP: like cantilever deformation start with 0 slope & displacm.

$W|_0 = 0$ No motion
 $\left. \frac{\partial W}{\partial x} \right|_0 = 0$ No Rotation
CLAMP @ $x=0$

$x=L$: PIN: condition about and other condition coming from forces equilibrium on dx @ L (using rotation equilibrium) → bending moment

$W|_L = 0$ No motion
 $M_y = EI \frac{\partial^2 W}{\partial x^2} \Big|_L = 0$
PIN @ $x=L$

from the dynamic vibration bending generc stationary solution:

$$W(x,t) = [A \cos(\gamma x) + B \sin(\gamma x) + C \cosh(\gamma x) + D \sinh(\gamma x)] \cos(\omega t + \varphi)$$

with $\gamma^4 = \omega^2 \frac{m}{EI}$

↓ plug it into the boundaries providing a system of 4 eq in 4 unk. LINEAR, OMG.

to write as $[H(\gamma L)] \underline{x} = 0$ $\underline{x} = (A \ B \ C \ D)^T$

preserving $\det([H]) = 0 \rightarrow$ find γ solution
 $\infty \ \gamma_k \ k=1, \dots$ solutions

and $\forall \gamma_k \rightarrow \omega_k \rightarrow \underline{x}_k$ non TRIVIAL solution

solvable

NUMERICALLY

by **MATLAB**

SCRIPTS

$\rightarrow$ write just the function

implementing matrix **H**

+ small matlab adaptation

EXAM

$\Downarrow$

finding the equations from

boundary conditions from $W(x,t)$

$$\frac{\partial W}{\partial x} = \gamma [-A \sin(\gamma x) + B \cos(\gamma x) + C \sinh(\gamma x) + D \cosh(\gamma x)] \cos(\omega t + \varphi)$$

$$\frac{\partial^2 W}{\partial x^2} = \gamma^2 [-A \cos(\gamma x) - B \sin(\gamma x) + C \cosh(\gamma x) + D \sinh(\gamma x)] \cos(\omega t + \varphi)$$

$\uparrow$ using this to implement ①, ②, ③, ④ and find the equations

simplify $\cos(\omega t + \varphi)$ because valid $\forall t$ boundaries

① $A + C = 0$

② $\gamma [B + D] = 0$

from here nothing about A, B, C, D can said...

linear equation with coeff. sin, cos...

③ $A \cos(\gamma L) + B \sin(\gamma L) + C \cosh(\gamma L) + D \sinh(\gamma L) = 0$

④ $-A \cos(\gamma L) - B \sin(\gamma L) + C \cosh(\gamma L) + D \sinh(\gamma L) = 0$

linear homogeneous equation

$$[H(\gamma L)] \underline{x} = 0 \quad \underline{x} = \begin{pmatrix} A \\ B \\ C \\ D \end{pmatrix}$$

on MATLAB we implement H matrix through function

$\Leftarrow 4 \times 4$ MATRIX

$$[H] = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ \cos(\gamma L) & \sin(\gamma L) & \cosh(\gamma L) & \sinh(\gamma L) \\ -\cos(\gamma L) & -\sin(\gamma L) & \cosh(\gamma L) & \sinh(\gamma L) \end{bmatrix}$$

once written $[H]$ on **MATLAB** we find **NUMERICAL SOLUTION**
through the main **MATLAB SCRIPT**



>> **vib_dps_20221103.m** SCRIPT

it uses a function "**clamphinge.m**" which implement $[H(xL)]$
our task is to find $[H]$ and implement it into a function
implement $[H]$

input of this function

provided during exam

$f_{MAX} [Hz] = ?$

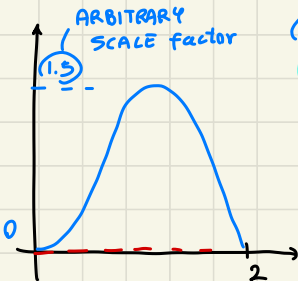
name func = 'clamphinge.m'

input the max frequency of analysis

$f_{MAX} = 400 Hz$

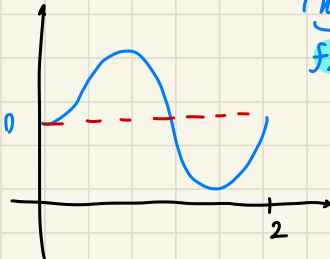
• focus 10:14 → parameters of problem beam
(to take from EXAM TEXT)

↓
{RUN}



$k=1$
 $f_1 = 102.11 Hz$

MODAL SHAPE



$k=2$
 $f_2 = 330.89 Hz$

@ $x=0$ 0 displace and slope as expected!

@ $x=L$ 0 displace as wanted

(and $\frac{\partial^2 w}{\partial x^2} \Big|_L = 0$)

↑ as fixed on boundaries

Respect

CANTILEVER beam

$f_1 \approx 25 Hz$, $f_2 \approx 30 Hz$

← increase on f_k

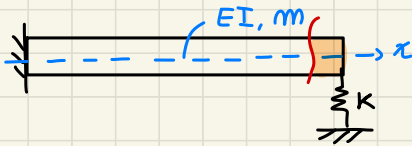
because of **STIFFER** beam
in this case!

ES 2 replace RIGID PIN WITH COUPED SPRING

We cover until now

CLAMP, FREE, PIN constrain → We can have a **SPRING**

NEW BOUNDARIES



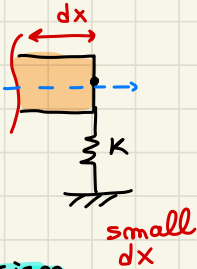
SO NOT a kinematic constrain BUT
an END subjected to **elastic force**
generated by the SPRING

@ $x=0$ ①, ② are same conditions

① $A + C = 0$

② $B + D = 0$

While @ $x=L$



ELASTICALLY supported end by spring (K)

NO KINEMATIC constrain (NOT displacement condition)

We presume dym equilibrium on dx piece of beam

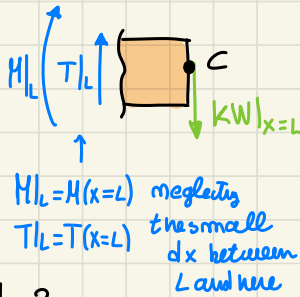
$$M = EI \frac{\partial^2 W}{\partial x^2}$$

$$T = \frac{\partial M}{\partial x} = EI \frac{\partial^3 W}{\partial x^3}$$

Recall **sigm**
convention



from dym
eq of this
piece of beam we find 2
equations:



$M_L = M(x=L)$ neglecting
 $T_L = T(x=L)$ the small
 dx between
 L and here

• Inertia force proportional
to $m \cdot dx \downarrow \approx 0$
small dx

• spring force proportional
to deformation of spring

$W(L, t) \leftarrow$ only elongation
of spring
 $\Delta l = W|_L$
 $W > 0$ elong
 $W < 0$ contract

• $\sum M_C = 0$ • no spring
contribution

③ • $T_L dx \approx 0 \Rightarrow M|_{x=L} = 0$

(same boundaries as PIN)

$$\frac{\partial^2 W}{\partial x^2} \Big|_L = 0$$

④ $\sum F_z = 0$ $\frac{EI \partial^3 W}{\partial x^3} \Big|_{x=L} - KW|_{x=L} = 0 \rightarrow \frac{EI \partial^3 W}{\partial x^3} \Big|_{x=L} - \frac{K}{EI} W|_{x=L} = 0$

from $\left. \frac{\partial^3 W}{\partial x^3} - \frac{K}{EI} W \right|_{x=L} = 0 \quad (4)$

↓
REWRITE LINEAR SYSTEM

from $W(x,t)$ we evaluate

$$\frac{\partial^3 W}{\partial x^3} = \gamma^3 [A \sin(\gamma x) - B \cos(\gamma x) + C \sinh(\gamma x) + D \cosh(\gamma x)] \cos(\omega t + \varphi)$$

① $A + C = 0$

② $B + D = 0$

③ $-A \cos(\gamma L) - B \sin(\gamma L) + C \cosh(\gamma L) + D \sinh(\gamma L) = 0$

④ $A \sin(\gamma L) - B \cos(\gamma L) + C \sinh(\gamma L) + D \cosh(\gamma L) - \frac{K}{EI} \gamma^3 [A \cos(\gamma L) + B \sin(\gamma L) + C \cosh(\gamma L) + D \sinh(\gamma L)] = 0$

↑
we can simplify $\cos(\omega t + \varphi)$, valid $\forall t$

↑
(linear homogeneous equation)

↑
the γ^3 before $\partial^3 W / \partial x^3$

$$[H] = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ -\cos(\gamma L) & -\sin(\gamma L) & \cosh(\gamma L) & \sinh(\gamma L) \\ \sin(\gamma L) - C_c \cos(\gamma L) & -\cos(\gamma L) - C_c \sin(\gamma L) & \sinh(\gamma L) - C_c \cosh(\gamma L) & \cosh(\gamma L) - C_c \sinh(\gamma L) \end{bmatrix}$$

$$C_c = K / (EI \gamma^3)$$

↑ new matrix expression to implement on **MATLAB**

through our function

⇒ **'clampspring'** = name func

↑
inside it I need to define also C_c

so found $\gamma = \gamma L / L$ define global variable
global EI L
global K

on the main and on "clampsprng.m"

we have global EI L

global K

$$\gamma = \gamma L / L;$$

$$C_c = K / EI / \gamma^3;$$

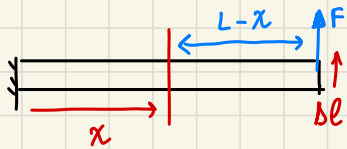
↙ $H = [\dots]$ implementation of H MATRIX

from here we use $\{A U U\}$ the script but

on the main we should define K value inside % Data

global K

we can find K from the problem



Cantilever beam + SPRING (K)



find K related to bending stiffness of CANTILEVER beam

found evaluating:

static force applied by us

$$\Delta \ell = W|_{x=L} \text{ under a given STATIC FORCE } (F) \rightarrow \text{STATIC CASE}$$

↑ displacement of W @ x=L

and on this case:

K related to beam stiffness can be

stiffness of cantilever beam

$$-(K) = F / \Delta \ell_w$$

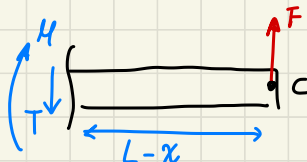
so we are doubling the K ELASTIC stiffness

The analysis of STATIC case is simple...

recalling

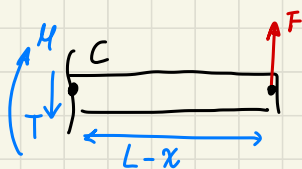
$$M = EI \frac{\partial^2 w}{\partial x^2} \text{ valid also on STATIC CASE!}$$

cutting the beam @ x



dynamic equilibrium

$$\sum M_c = 0 \Rightarrow M = F(L-x)$$



but $M = \frac{\cancel{EI}}{\cancel{EI}} \frac{\partial^2 W}{\partial x^2} = F(L-x)$

so we express $\partial^2 W / \partial x^2$

$$\frac{\partial^2 W}{\partial x^2} = \frac{F}{EI} (L-x)$$

transversal displacement for different sections of the beam on static case
With STAIL FORLEF

integrating 2 time

$$\frac{\partial W}{\partial x} = \frac{F}{EI} \left(Lx - \frac{x^2}{2} \right) + C_1$$

constant integration

$$W(x) = \frac{F}{EI} \left(L \frac{x^2}{2} - \frac{x^3}{6} \right) + \cancel{C_1} x + \cancel{C_2}$$

W displacement $W(x)$
When F applied
STATIC

presumption of static

case @ $x=0$ $W(0) = 0$ NO DISPLACE $\rightarrow C_2 = 0$

$\frac{\partial W}{\partial x}(0) = 0$ NO ROTATION $\rightarrow C_1 = 0$

and evaluating

$$W(x) = \frac{F}{EI} \left(L \frac{x^2}{2} - \frac{x^3}{6} \right) \quad \Delta l = W(L) = \frac{F}{EI} \left(\frac{L^3}{2} - \frac{L^3}{6} \right) =$$

so because equivalent

bending stiffness of cantilever

beam loaded by F is

$$K = \frac{F}{\Delta l} = \frac{3EI}{L^3} \quad [N/m] \quad (\text{OK})$$

so setting

$$K = \frac{3EI}{L^3}$$

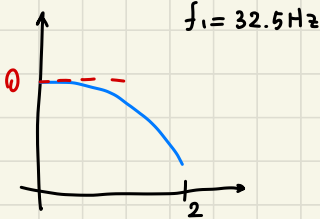
IT'S good to set STIFFNESS to same stiffness of cantilever when statically loaded on free end transversal

so stiffness K of lumped beam
related to the cantilever behav

on MATLAB $K = 3 * EI / L^3$;

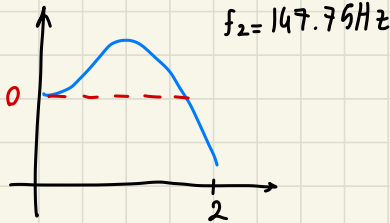
{RUN} the code of main...

$f_{MAX} = 400 \text{ Hz}$



lower f_k respect
PIN case

but low stiff K spring

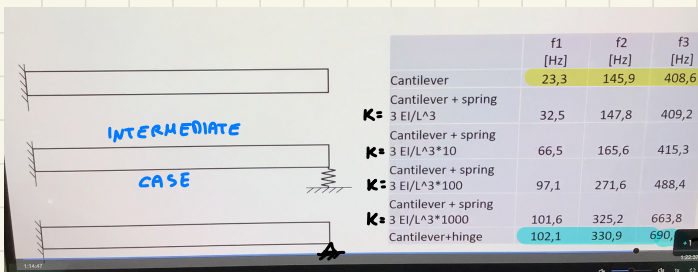


We are between the
PIN OR FREE case

low K STIFFNESS
between FREE $x=L$ (0 STIFF)
and PIN $x=L$ (∞ STIFF) } NOW SPRING (K STIFF)

We get f_k between
free case and PIN case !

if $K \uparrow \uparrow$ we
move near
to CLAMP-PIN case

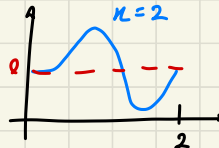
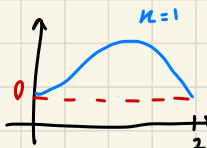


∞ STIFF

increasing of
nat freq for ← intermediate
increasing case
STIFFNESS

00 STIFF

if $K \uparrow \uparrow$



the displacement @ $x=L$
is near 0, nearly as STIFF
as an ∞ STIFF PIN

$K \uparrow$ stiffer beam

for the small K instead we get a big neg
displacement @ $x=L$ MEAN PURE CANTILEVER

in fact if $K=0$ we get mean CLAMP-FREE
case, pure CANTILEVER

similar modal shape
lower natural freq of low stiffness